

Nyasha Hama

Backend & Platform Engineer | Go, PostgreSQL, Redis, Rust | APIs, Data & Infrastructure

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Summary

Backend and platform engineer building public, testable systems around Go APIs, PostgreSQL data models, Redis-backed services, Rust infrastructure components, and TypeScript product surfaces.

Public repositories show auth/RBAC, migrations, pgx/sqlc, Redis/NATS, Dockerized local stacks, CI workflows, health/readiness endpoints, Prometheus metrics, and live demos.

Skills

Languages: Go, Rust, TypeScript, JavaScript/Node.js, Java, C++, C#, SQL

Backend & Platform: REST APIs, service design, JWT authentication, role-based access control, background jobs, caching, messaging

Data: PostgreSQL, Redis, pgx, sqlc, schema design, query optimization, indexing

Infrastructure & Tooling: Docker, Linux, Prometheus, structured logging, GitHub Actions, CI/CD

Experience

Backend & Platform Engineer, Independent Software Engineer – Remote Jan 2023 – present

- Built independent systems spanning property operations, healthcare access, risk workflows, developer tooling, and infrastructure-oriented services.
- Designed Go services with JWT/RBAC, PostgreSQL schema design, migrations, pgx/sqlc, Redis caching, observability, CI checks, and Dockerized local environments.

Projects

StrataHQ

Property-management platform for South African sectional-title schemes covering levy operations, maintenance workflows, AGM administration, communications, and document access.

- Built a Next.js and Go monorepo with scheme-scoped workflows for agents, trustees, and residents.
- Modeled auth, scheme management, levy operations, maintenance, and billing with PostgreSQL, pgx, sqlc, goose migrations, Redis, Stripe, email, tests/CI, and deployment paths.
- Links: [GitHub](#) | [Live](#)

Healthcare Access Connector

Healthcare access backend connecting patients and providers through scheduling, search, notifications, and role-aware workflows.

- Designed patient, provider, and appointment flows with PostgreSQL-backed domain modeling, validation, and explicit service boundaries.
- Implemented JWT/RBAC, rate limiting, Redis caching, NATS messaging, Docker-based local infrastructure, Prometheus metrics, structured logging, health endpoints, and graceful shutdown.
- Links: [GitHub](#)

go-backend-scaffold

Reusable Go backend scaffold for starting production-minded API services with auth, migrations, observability, and CI quality gates.

- Built a reusable Go service foundation around chi, pgx, sqlc, goose migrations, JWT auth, Redis, Prometheus, slog, Docker, and GitHub Actions.
- Added release-readiness checks for migration verification, linting, race-tested CI runs, Docker builds, vulnerability triage, clean-path bootstrap, and adopter documentation.
- Links: [GitHub](#)

TitleChain

Property transaction intelligence platform for South African conveyancing, banking, and insurance workflows.

- Built a monorepo foundation spanning a Next.js portal, Go API/control plane, Rust data-processing worker, PostgreSQL migrations, and Docker infrastructure.
- Added case workflow, property projection, source-ingestion, decision-engine, and pilot-customer verification paths for historically versioned property intelligence.
- Links: [GitHub](#) | [Live](#)

Additional Projects

- guard-rail | Rust policy-enforcement runtime with readiness, metrics, audit persistence, and pilot deployment docs | [GitHub](#)
- Burnout Predictor | Go and Next.js workload-analysis product for cognitive-load scoring and forecasts | [GitHub](#) | [Live](#)
- AgencyForge | AI-assisted campaign generation platform | [GitHub](#)
- FormFlow AI | Government-form automation workflow with document parsing and auditability | [GitHub](#)

Education

University of Johannesburg — BSc Computer Science Coursework, completed semesters through final-year coursework, 2022–2024